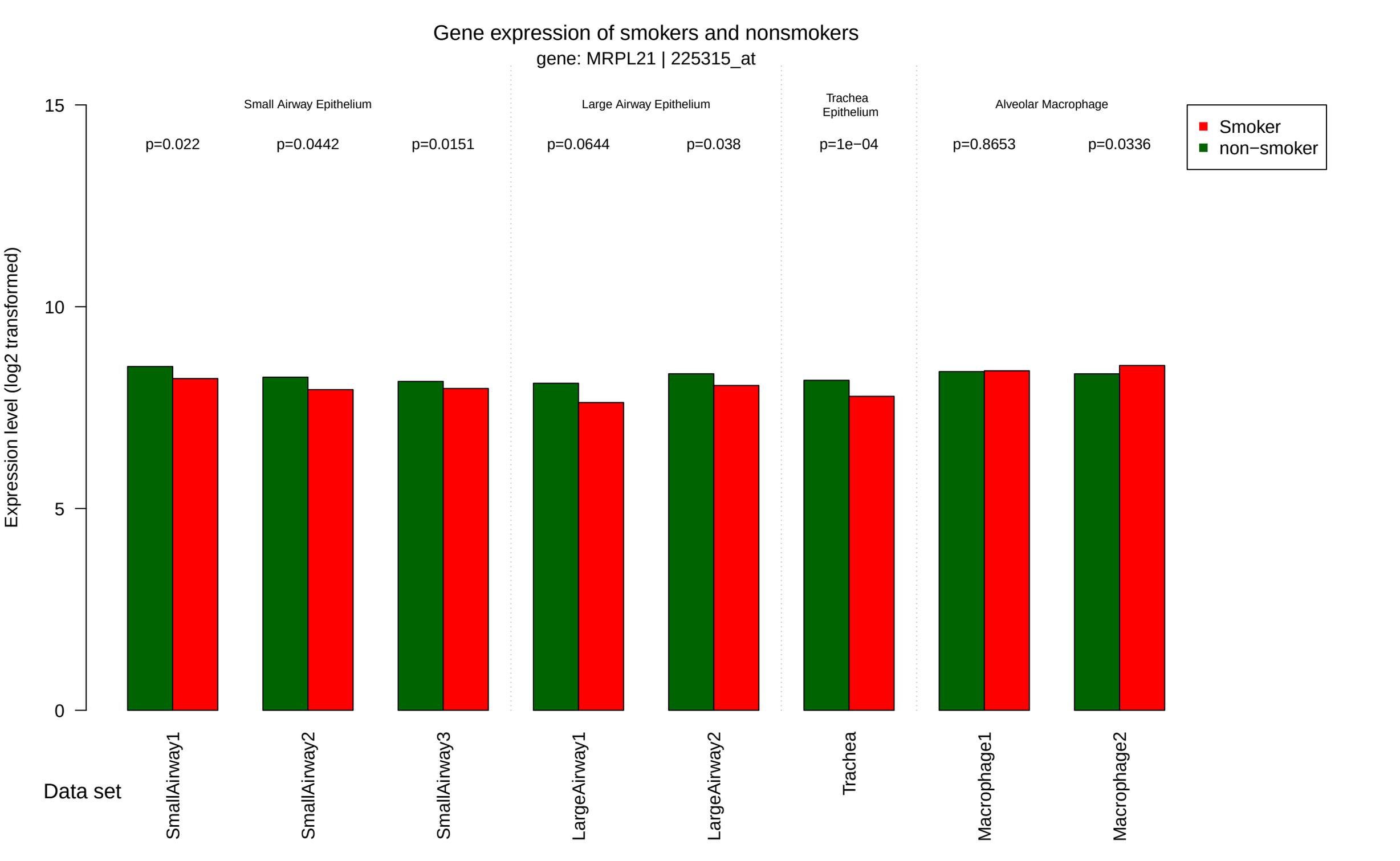
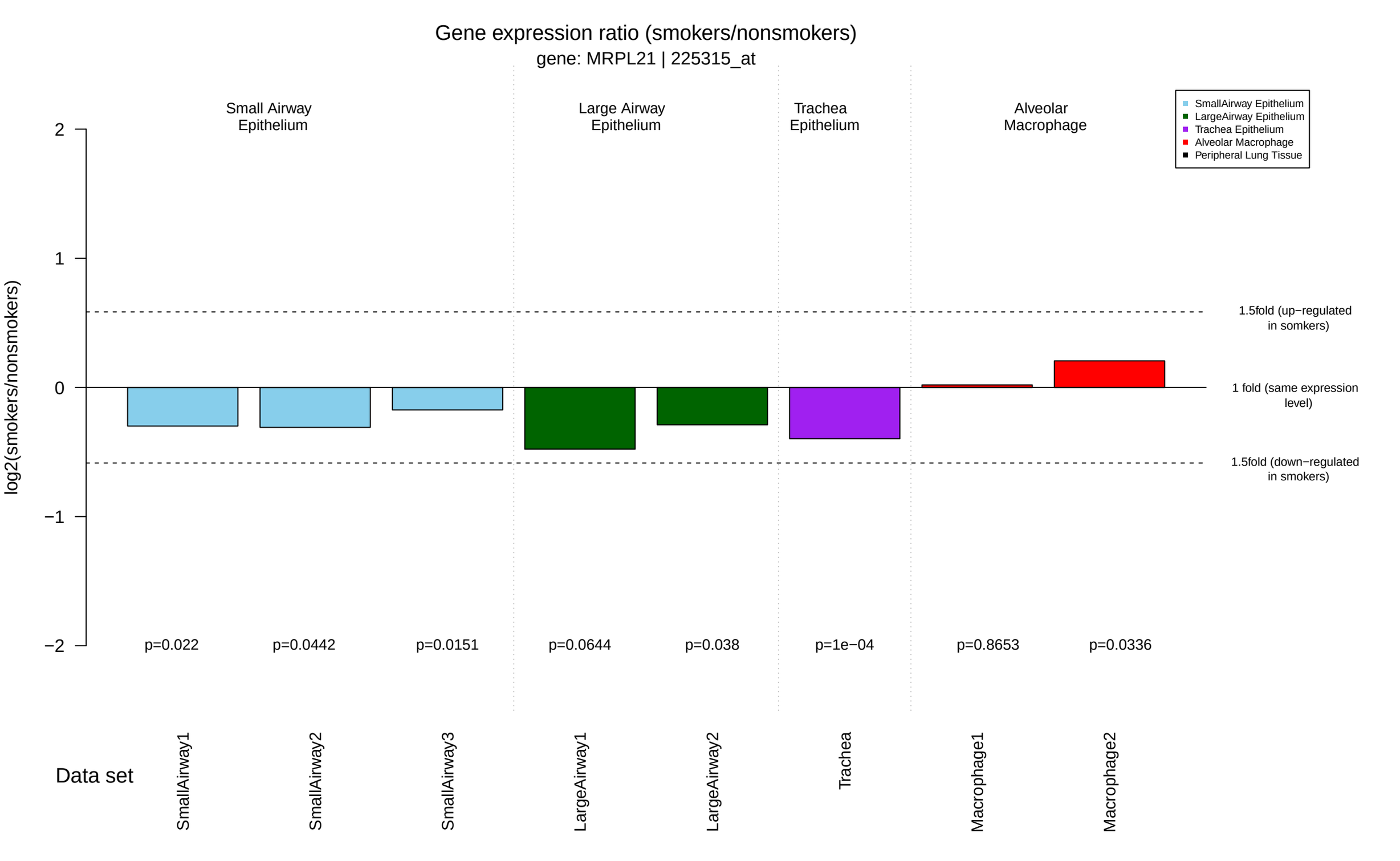
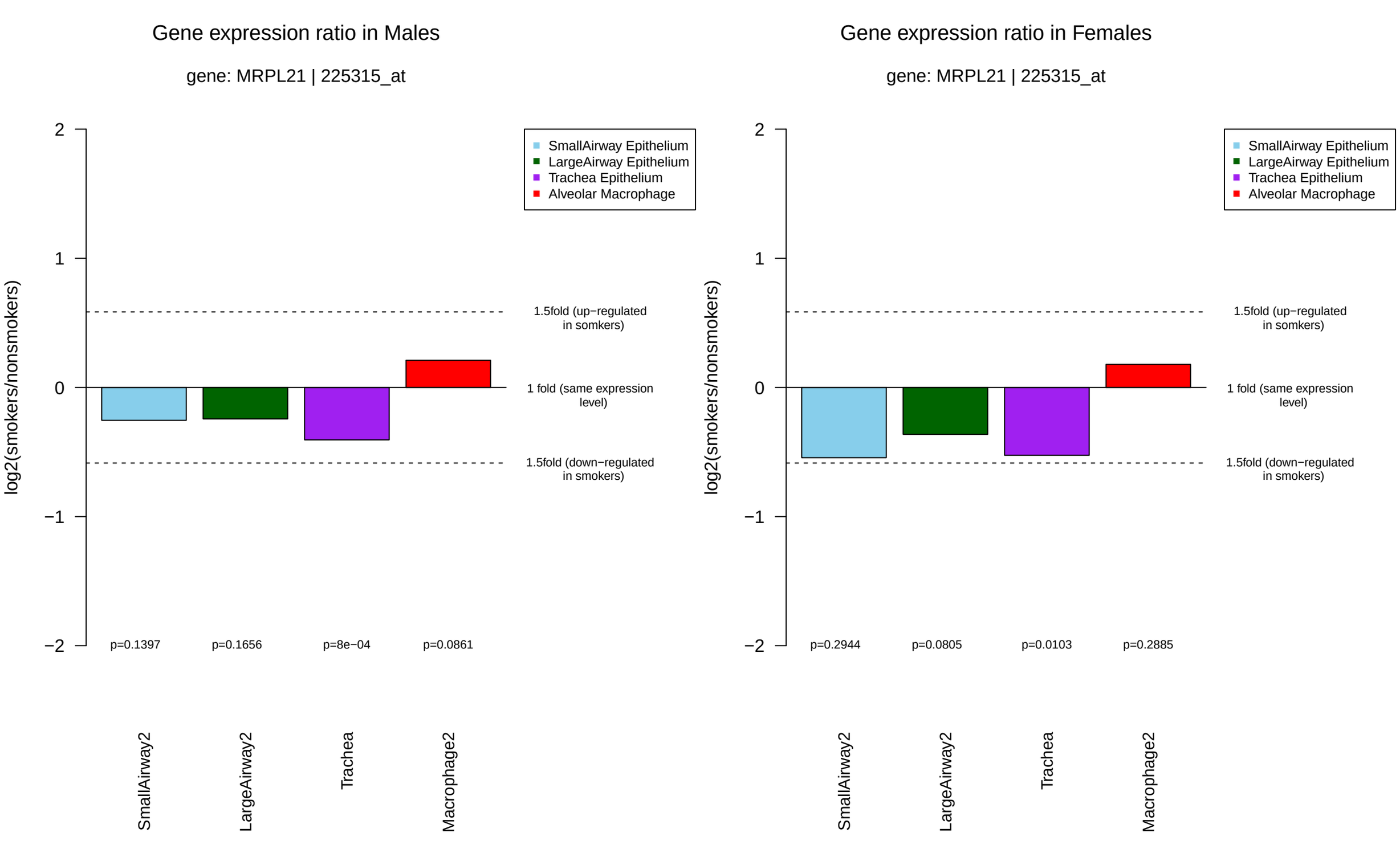
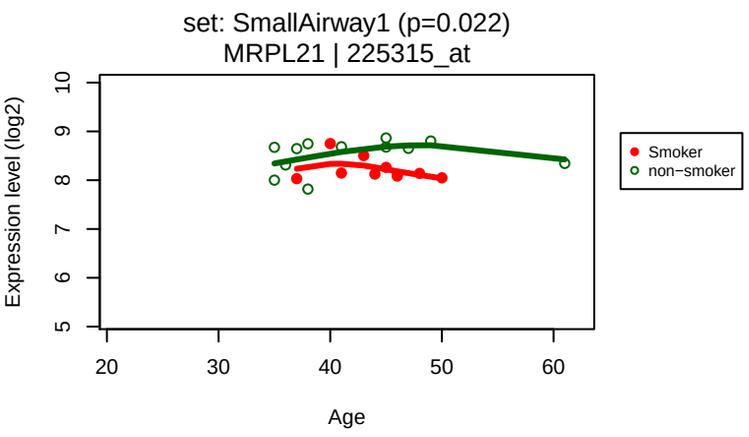
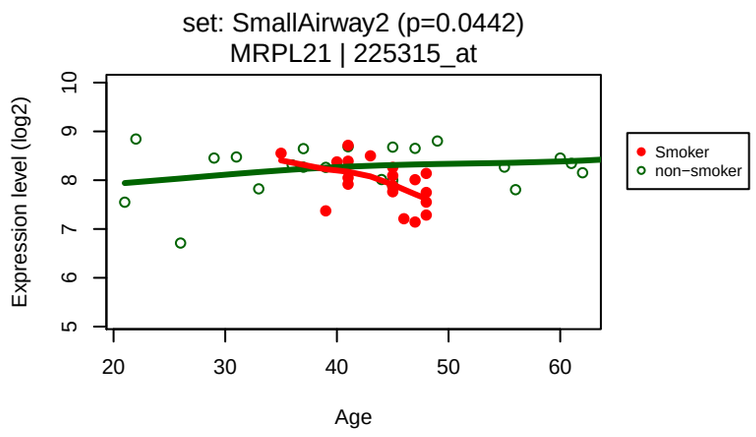
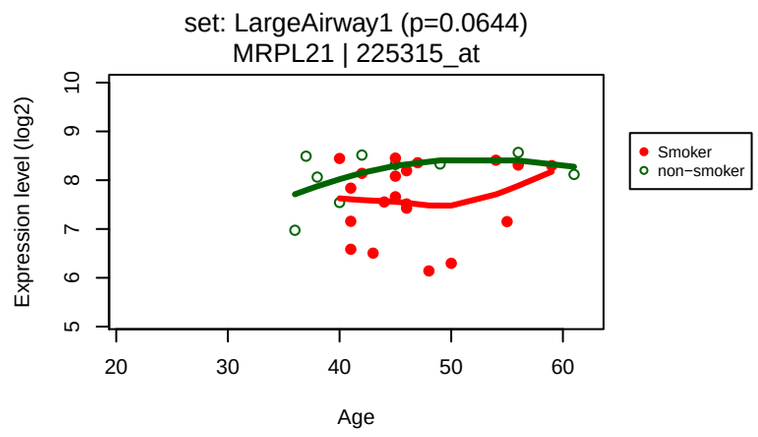

Gene expression of smokers and nonsmokers
gene: MRPL21 | 225315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.022
p=0.0442
p=0.0151
p=0.0644
p=0.038
p=1e−04
p=0.8653
p=0.0336
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

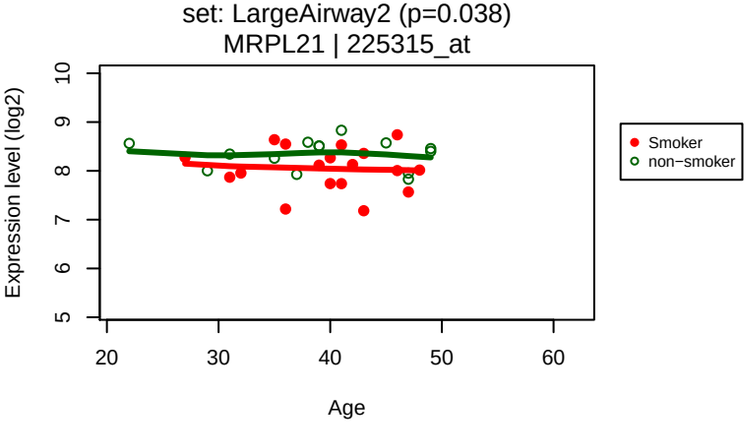

Gene expression ratio (smokers/nonsmokers)
gene: MRPL21 | 225315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.022
p=0.0442
p=0.0151
p=0.0644
p=0.038
p=1e−04
p=0.8653
p=0.0336
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

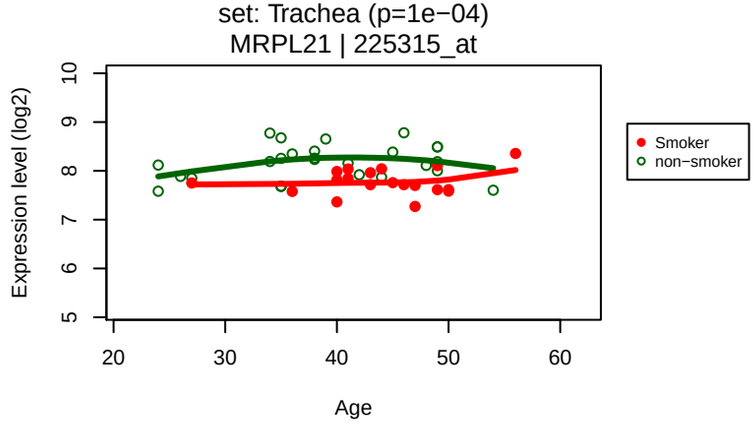

Gene expression ratio in Males
gene: MRPL21 | 225315_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1397
p=0.1656
p=8e−04
p=0.0861
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL21 | 225315_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2944
p=0.0805
p=0.0103
p=0.2885
SmallAirway2
LargeAirway2
Trachea
Macrophage2

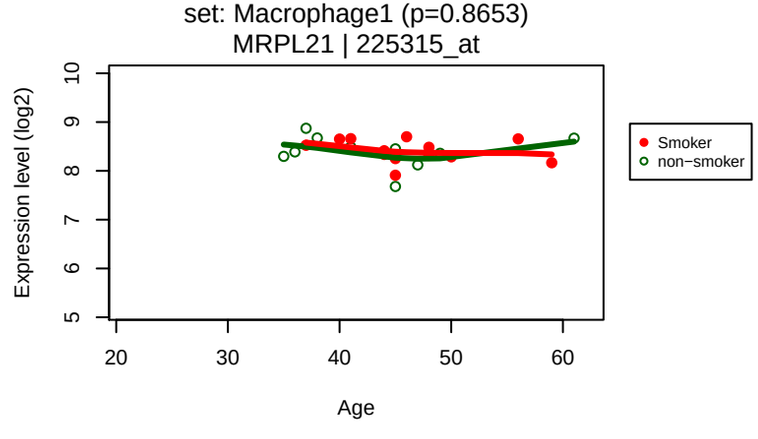

set: SmallAirway1 (p=0.022)
MRPL21 | 225315_at
Expression level (log2)
Age
Smoker
non−smoker

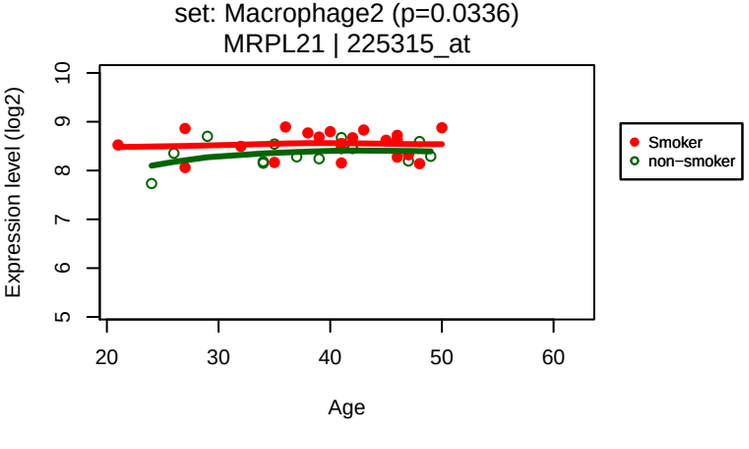

set: SmallAirway2 (p=0.0442)
MRPL21 | 225315_at
Expression level (log2)
Age
Smoker
non−smoker

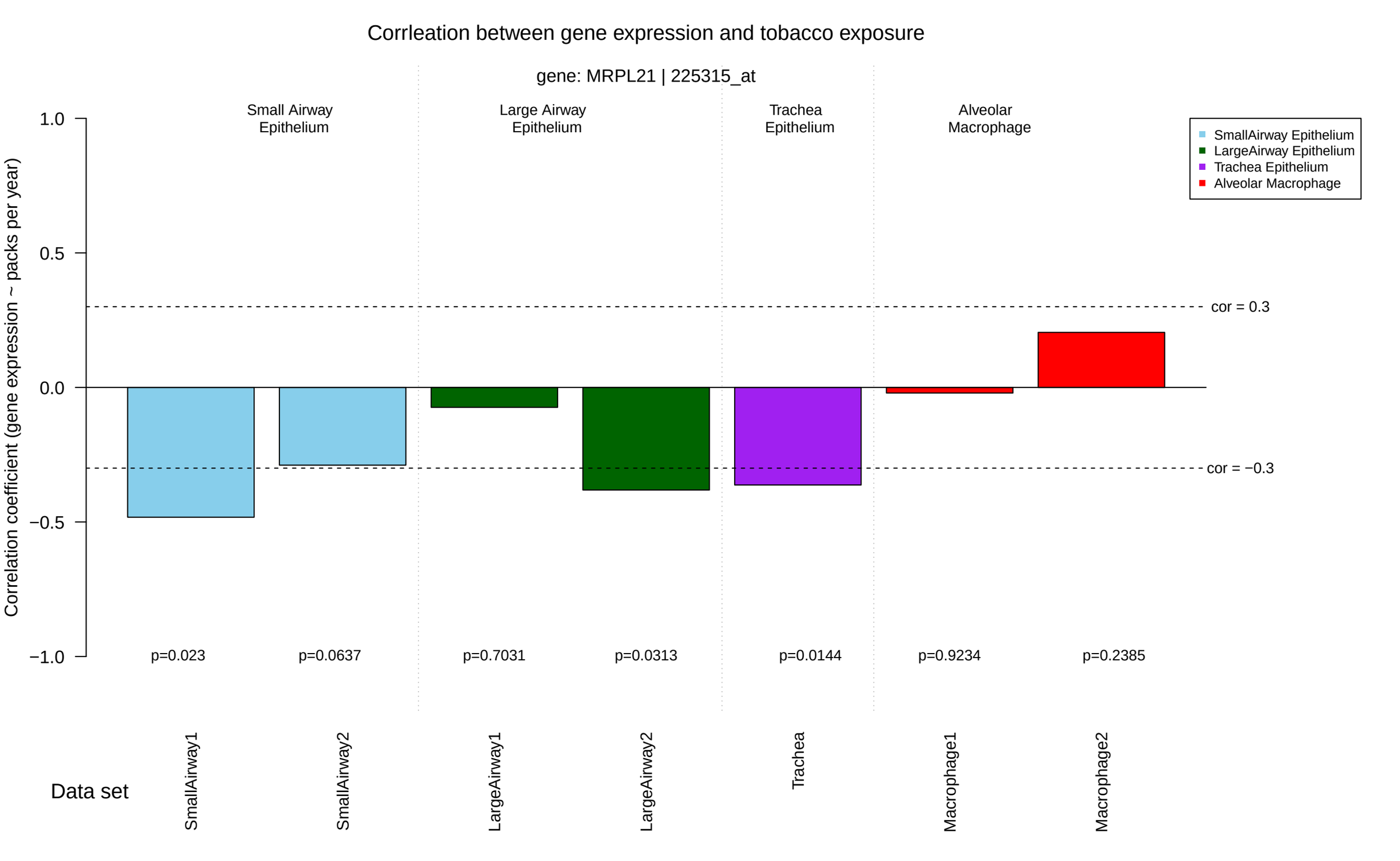

set: LargeAirway1 (p=0.0644)
MRPL21 | 225315_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.038)
MRPL21 | 225315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
MRPL21 | 225315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8653)
MRPL21 | 225315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0336)
MRPL21 | 225315_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL21 | 225315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.023
p=0.0637
p=0.7031
p=0.0313
p=0.0144
p=0.9234
p=0.2385
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2